

CA-C30L: Mobile Application Development Lab

LAB PROGRAM-6 : UI with all VIEWS

6. Create an UI with all views.

Program description :

- The program displays the various views used in android and displays different toast messages with respect to the view clicked

Steps :

- [Use Linear layout] Create the activity.xml file by defining all the view tags for Button, Image Button, Checkbox, Radio Button and toggle button. Define their ids, and include text in string.xml file for all.
- Use a drawable icon resource for image button (min 48 X48 dp)
- Declare objects of all the views and find them by their id
- Define the listener methods for all the views (methods for image button and button are similar ,while those for checkbox, radiobutton and toggle are similar)

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/
android"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity"
android:orientation="vertical">

    <TextView
```

```
        android:id="@+id/textView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="@string/heading" />
<Button
    android:id="@+id/buttonExample"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="@string/example_button" />
<CheckBox
    android:id="@+id/checkboxExample"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="@string/checkbox_example" />
<ToggleButton
    android:id="@+id/toggleButtonExample"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"

    android:text="@string/togglebutton_example"
    android:textOn="@string/on"
    android:textOff="@string/off"
    />
<RadioGroup
    android:id="@+id/radioGroupExample"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content" >
    <RadioButton
        android:id="@+id/radioBtn1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:checked="true"
        android:text="@string/radiobutton_1"
    />

    <RadioButton
        android:id="@+id/radioBtn2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/radiobutton_2"
    />
```

```
        <RadioButton
            android:id="@+id/radioBtn3"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="@string/radiobutton_3"
        />
    </RadioGroup>
    <ImageButton
        android:id="@+id/imageButtonExample"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"

        android:src="@drawable/baseline_add_ic_call_24"
        android:contentDescription="@string/todo"
    />

</LinearLayout>
```

MainActivity.java

```
package com.example.uidemo;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.CompoundButton;
import android.widget.ImageButton;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.Toast;
import android.widget.ToggleButton;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends
AppCompatActivity {
    CheckBox exampleCheckBox;
```

```
@Override
protected void onCreate(Bundle
savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    //Declaring view objects and finding them
    by their id
    Button example =
    findViewById(R.id.buttonExample);
    exampleCheckBox =
    findViewById(R.id.checkBoxExample);
    RadioGroup exampleRadioGroup =
    findViewById(R.id.radioGroupExample);
    ToggleButton exampleToggleButton =
    findViewById(R.id.toggleButtonExample);
    ImageButton exampleImageButton =
    findViewById(R.id.imageButtonExample);

    //Button Listener method implementation
    example.setOnClickListener(new
    View.OnClickListener() {
        @Override
        public void onClick(View v) {

            Toast.makeText(getApplicationContext(), "Button
            clicked", Toast.LENGTH_SHORT).show();
        }
    });
    //Image Button Listener method
    implementation
    exampleImageButton.setOnClickListener(new
    View.OnClickListener() {
        @Override
        public void onClick(View v) {

            Toast.makeText(getApplicationContext(), "Calling
            Image button", Toast.LENGTH_SHORT).show();
        }
    });
    //Toggle Button Listener method implementation
```

```
exampleToggleButton.setOnCheckedChangeListener(new
CompoundButton.OnCheckedChangeListener() {
    @Override
    public void onCheckedChanged(CompoundButton
buttonView, boolean isChecked) {

        if(exampleToggleButton.isChecked()) {

            Toast.makeText(getApplicationContext(), "ON Button
clicked", Toast.LENGTH_SHORT).show();

        }

        else {

            Toast.makeText(getApplicationContext(), "OFF
Button clicked", Toast.LENGTH_SHORT).show();

        }

    }

});
//CheckBox Listener method implementation
```

```
exampleCheckBox.setOnCheckedChangeListener(new
CompoundButton.OnCheckedChangeListener() {
    @Override
    public void onCheckedChanged(CompoundButton
buttonView, boolean isChecked) {
        if (isChecked) {

            Toast.makeText(getApplicationContext(), "Checked",
Toast.LENGTH_SHORT).show();

        }
        else{

            Toast.makeText(getApplicationContext(),
"Unchecked", Toast.LENGTH_SHORT).show();

        }

    }

});
```

```
//Radio Button Listener method implementation

exampleRadioGroup.setOnCheckedChangeListener(new
RadioGroup.OnCheckedChangeListener() {
    @Override
    public void
onCheckedChanged(RadioGroup group, int
checkedButtonId) {
    if (checkedButtonId== R.id.radioBtn1) {

Toast.makeText(getApplicationContext(), "Course
selected-Arts", Toast.LENGTH_SHORT).show();
    }
    if (checkedButtonId == R.id.radioBtn2 ) {

Toast.makeText(getApplicationContext(), " Course
selected-Science ", Toast.LENGTH_SHORT).show();
    }
    if(checkedButtonId== R.id.radioBtn3 ) {

Toast.makeText(getApplicationContext(), " Course
selected-Commerce", Toast.LENGTH_SHORT).show();
    }
    }

    });

}

}

****
```